ANSWER KEY

Unit 5: The Pythagorean Theorem — All Sections

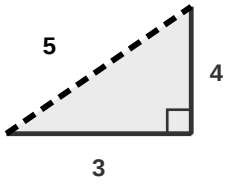
Section 5.1 — Introduction & How to Use

Q1. c = 5

$3^2+4^2 = 9+16 = 25$

$c = \sqrt{25} = 5$

→ *3-4-5 triple.*

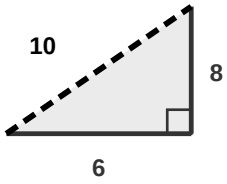


Q2. c = 10

$6^2+8^2 = 36+64 = 100$

$c = \sqrt{100} = 10$

→ *3-4-5 ×2.*

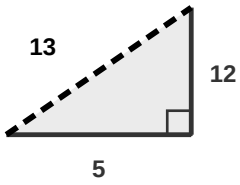


Q3. c = 13

$5^2+12^2 = 25+144 = 169$

$c = \sqrt{169} = 13$

→ *5-12-13 triple.*

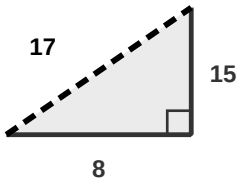


Q4. c = 17

$8^2+15^2 = 64+225 = 289$

$c = \sqrt{289} = 17$

→ *8-15-17 triple.*



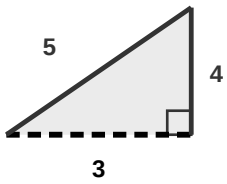
Q5. a = 3

$a^2+4^2 = 5^2$

$a^2 = 25-16 = 9$

$a = \sqrt{9} = 3$

→ *Subtract leg² from hyp².*



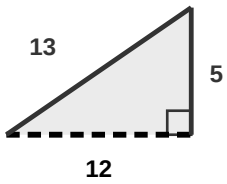
Q6. a = 12

$a^2+5^2 = 13^2$

$a^2 = 169-25 = 144$

$a = \sqrt{144} = 12$

→ *5-12-13 triple.*



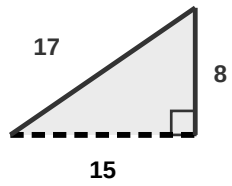
Q7. a = 15

$a^2+8^2 = 17^2$

$a^2 = 289-64 = 225$

$a = \sqrt{225} = 15$

→ 8-15-17 triple.



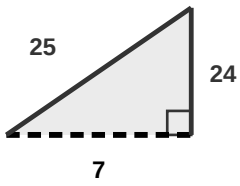
Q8. a = 7

$a^2+24^2 = 25^2$

$a^2 = 625-576 = 49$

$a = \sqrt{49} = 7$

→ 7-24-25 triple.

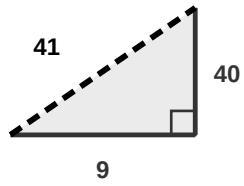


Q9. c = 41

$9^2+40^2 = 81+1600 = 1681$

$c = \sqrt{1681} = 41$

→ 9-40-41 triple.



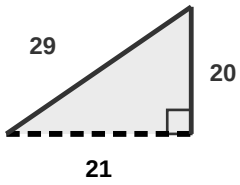
Q10. a = 21

$a^2+20^2 = 29^2$

$a^2 = 841-400 = 441$

$a = \sqrt{441} = 21$

→ 20-21-29 triple.

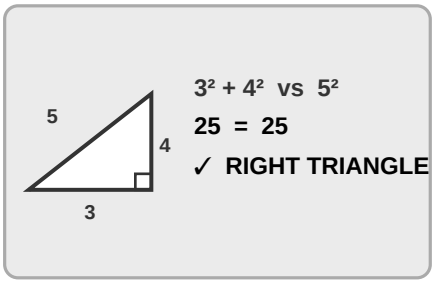


Section 5.2 — Converse

Q1. YES

$3^2+4^2=25$, $5^2=25$ ✓

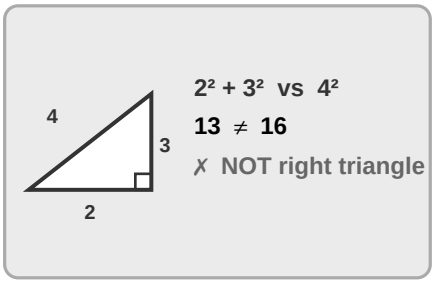
→ 3-4-5 triple.



Q2. NO

$2^2+3^2=13$, $4^2=16$ ✗ $(13\neq16)$

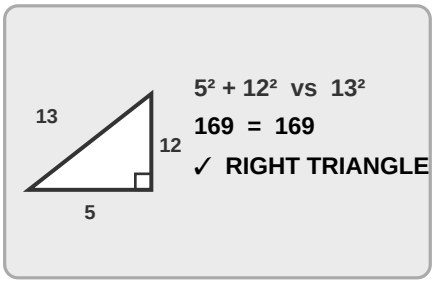
→ Not a triple.



Q3. YES

$5^2+12^2=169$, $13^2=169$ ✓

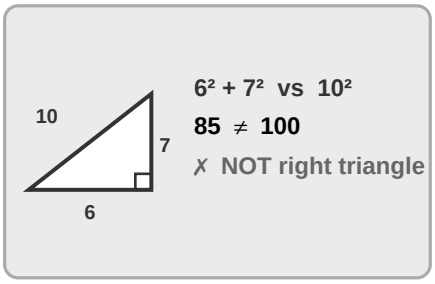
→ 5-12-13 triple.



Q4. NO

$6^2+7^2=85$, $10^2=100$ ✗

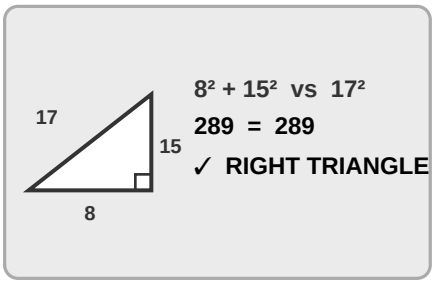
→ 85≠100.



Q5. YES

$8^2+15^2=289,\ 17^2=289$ ✓

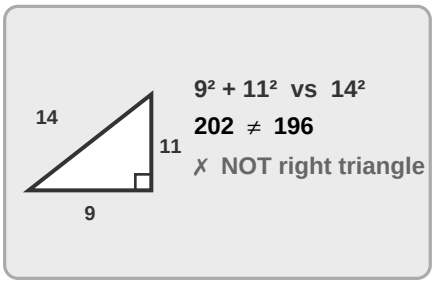
→ *8-15-17 triple.*



Q6. NO

$9^2+11^2=202,\ 14^2=196$ ✗

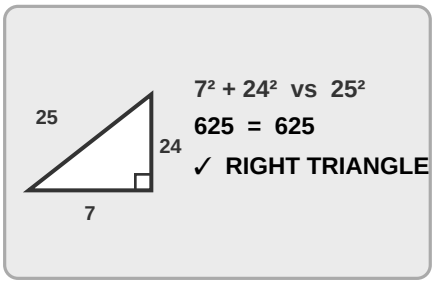
→ *202≠196.*



Q7. YES

$7^2+24^2=625,\ 25^2=625$ ✓

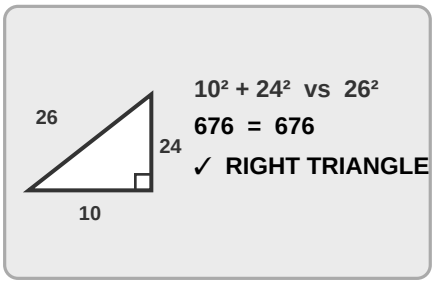
→ *7-24-25 triple.*



Q8. YES

$10^2+24^2=676,\ 26^2=676$ ✓

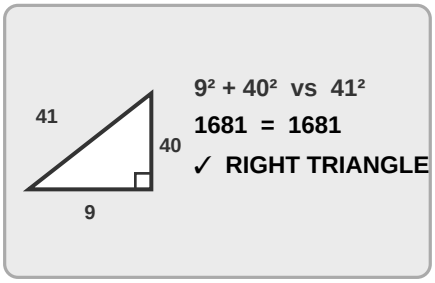
→ *5-12-13 ×2.*



Q9. YES

$9^2+40^2=1681,\ 41^2=1681$ ✓

→ *9-40-41 triple.*

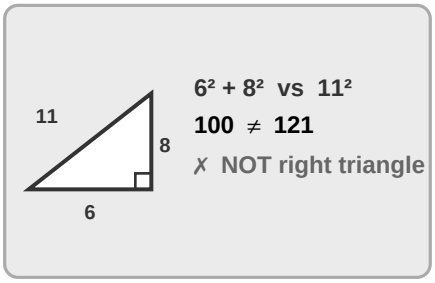


Q10. NO

$6^2+8^2=100,\ 11^2=121$ ✗

(6-8-10 works, not 6-8-11!)

→ *Trick!*



Section 5.3 — Word Problems

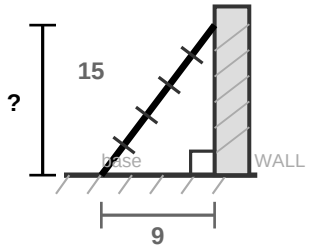
Q1. h = 12 ft

$9^2+h^2=15^2$

$81+h^2=225$

$h^2=144 \rightarrow h=12$

→ *3-4-5×3. Ladder=hyp.*

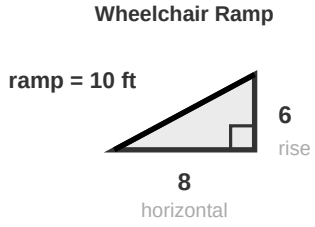


Q2. 8 ft

$a^2+6^2=10^2$

$a^2=64 \rightarrow a=8$

→ *6-8-10 triple.*

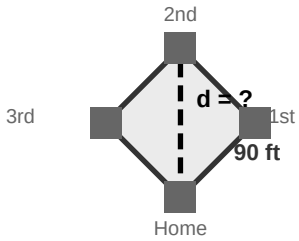


Q3. ≈ 127 ft

$d^2=90^2+90^2=16200$

$d=\sqrt{16200}\approx 127.3$

→ *Round to the nearest foot.*

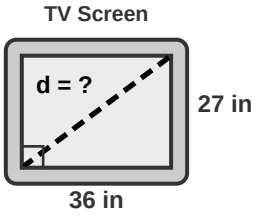


Q4. 45 in

$36^2+27^2=1296+729=2025$

$d=\sqrt{2025}=45$

→ *3-4-5×9.*

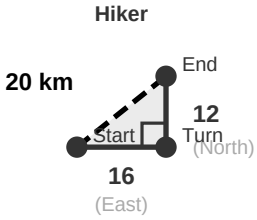


Q5. 20 km

$16^2+12^2=256+144=400$

$d=\sqrt{400}=20$

→ *3-4-5×4.*

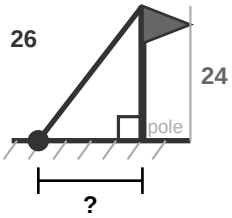


Q6. 10 ft

$a^2+24^2=26^2$

$a^2=676-576=100 \rightarrow a=10$

→ *5-12-13×2.*

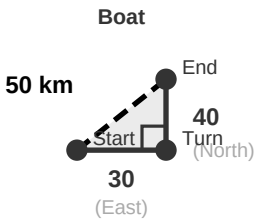


Q7. 50 km

$30^2+40^2=900+1600=2500$

$d=\sqrt{2500}=50$

→ *3-4-5×10.*

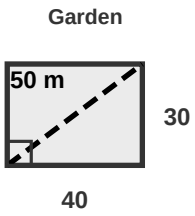


Q8. 50 m

$40^2+30^2=1600+900=2500$

$d=\sqrt{2500}=50$

→ *3-4-5×10.*



Section 5.4 — Coordinate Plane Distance

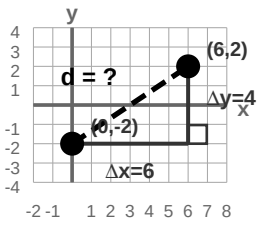
Q1. d=√52≈7.21

Δx=6, Δy=4

$d^2=36+16=52$

d=√52≈7.21

→ *Not a perfect square, so round.*



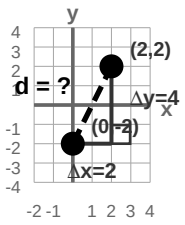
Q2. d=√20≈4.47

Δx=2, Δy=4

$d^2=4+16=20$

d=√20≈4.47

→ *Not a perfect square, so round.*



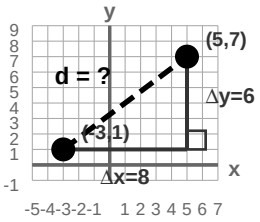
Q3. d = 10

Δx=8, Δy=6

$d^2=64+36=100$

d=10

→ *6-8-10 triple!*



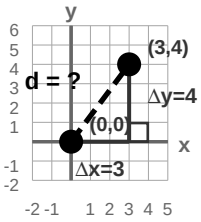
Q4. d = 5

Δx=3, Δy=4

$d^2=9+16=25$

d=5

→ *3-4-5 triple.*



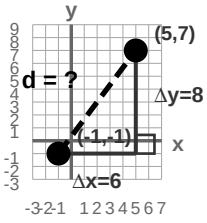
Q5. d = 10

Δx=5−(−1)=6, Δy=7−(−1)=8

$d^2=36+64=100$

d=10

→ *Subtract negatives carefully. 6-8-10 triple.*



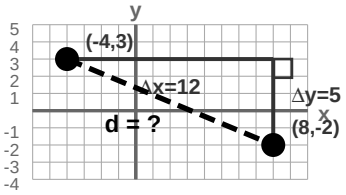
Q6. d = 13

Δx=8−(−4)=12, Δy=|−2−3|=5

$d^2=144+25=169$

d=13

→ *5-12-13 triple.*



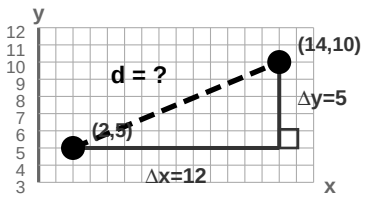
Q7. 13 miles

$\Delta x=12,\ \Delta y=5$

$d^2=144+25=169$

$d=13$ miles

→ 5-12-13 triple.



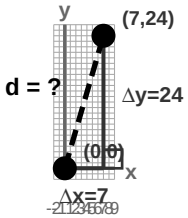
Q8. d = 25

$\Delta x=7,\ \Delta y=24$

$d^2=49+576=625$

$d=25$

→ 7-24-25 triple.

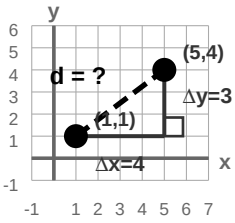


Q9. AC = 5

AB=4 (horiz), BC=3 (vert)

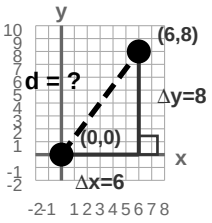
$AC^2=16+9=25 \rightarrow AC=5$

→ 3-4-5 triple.



Q10. PR = 10

PQ=6, QR=8

$PR^2=36+64=100 \rightarrow PR=10$

→ Right angle at Q(6,0).